Reverse Engineering (ROL) - Exam: May 2024

FINAL EXAM

NAME: .... Group: ....

Time to complete the final exam: **90 minute**. Minimum points to pass the final exam: **18p**. Good luck!

1. **Good usage of reverse engineering**

   Which of the following represent proper/good/ethical usage of reverse engineering (e.g., in terms of ethical hacking)? Mark your selection with OK. Mark the rest with NOK. The first one is done as an example.

   (a) Bypass the authentication of a paid app. ... **NOK** ...
   (b) Check for code errors. **(2p)** ....
   (c) Modify the code to get access to functionalities you should normally not be entitled to. **(2p)** ....
   (d) Check for undocumented functionalities. **(2p)** ....
   (e) Check for behaviour in special cases, boundary conditions, etc. **(2p)** ....
   (f) Crack an app from its free version to its paid version **(2p)** ....
   (g) Search the code for possible malware. **(2p)** ....

2. **General concepts/notions**

   Choose the correct concept/notion (cut the incorrect answers with a line) or fill in the blanks in the text below. The first one is done as an example.

   Mathew is a reverse engineer. He is looking into ~~*open-souce*~~/*non open-source* software to find its functionality and search for possible malware. In his job, he gets access to some *binary files*/*text files* **(2p)**, such as, e.g., *PE*/*TXT* **(2p)** files. He is currently examining an ELF file, identified as such by the file signature (or magic number) 7F 45 4C 46/50 45 00 00 **(2p)**. To perform his tasks, he is initially examining the code without running it, which means he is performing a *static analysis*/*dynamic analysis* **(2p)**. Mathew runs the .... **(2p)** command to display information about the file; he finds the start of the section header to be 0x4AAED0 and the size of the section headers to be 64 bytes. He further uses the option –section-headers of the same command and finds the index of section .data to be 23. Mathew computes that the .data section starts at .... **(2p)**. In the .data section he is identifying the sequence (0x) 36 32 37 34, which he immediately understands it must be a *number value*/*cmp instruction* **(2p)**. Mathew is further performing a more in-depth analysis and debugging. He is using gdb-peda and runs *vmmap*/*aslr* **(2p)** to find the virtual mapping address ranges ...

3. **Assembly**

   Given the assembly code below, explain its functionality in 1-2 phrases. Explicitly mention the value of a "secret value" of interest and what it represents. **(8p)**

```
vrf_pin(int, int):
        push    rbp
        mov     rbp, rsp
        mov     DWORD PTR [rbp-4], edi
        mov     DWORD PTR [rbp-8], esi
        mov     eax, DWORD PTR [rbp-4]
        cmp     eax, DWORD PTR [rbp-8]
        sete    al
        movzx   eax, al
        pop     rbp
        ret
main:
        push    rbp
        mov     rbp, rsp
        sub     rsp, 32
        mov     DWORD PTR [rbp-4], 7253
        movabs  rax, 7598805555174203734
        movabs  rdx, 5282757987228741231
        mov     QWORD PTR [rbp-32], rax
        mov     QWORD PTR [rbp-24], rdx
        mov     WORD PTR [rbp-16], 78
        mov     edx, DWORD PTR [rbp-4]
        mov     eax, DWORD PTR [rbp-8]
        mov     esi, edx
        mov     edi, eax
        call    vrf_pin(int, int)
        mov     DWORD PTR [rbp-12], eax
        mov     eax, 0
        leave
        ret
```

Answer:

4. **True/False**

Respond with true (T) or false (F) and give a short explanation of your choice.

(a) ASLR allows segment addresses to change while the process is running. **(2p)**
Answer:

(b) GOT and PLT segments exist for both statically and dynamically linked ELF binaries. **(2p)**
Answer:

TOTAL available: 40p